12TH CHEMISTRY · CHAPTER 25 · FULL CHAPTER MCQS

Organic Synthesis

119 topic-wise MCQs from the complete chapter with answers and reasons

INTRODUCTION

1. The study of organic chemistry is organised around:

(a) functional groups (b) molecular masses
(c) boiling points (d) isotopes

Answer: (a) functional groups — Each functional group defines a family of organic compounds.

2. The part of an organic compound where most of the reactions occur is called its:

(a) alkyl group (b) carbon skeleton
(c) functional group (d) synthon

Answer: (c) functional group — This is the definition of a functional group.

3. Designing possible ways to build a target molecule from simpler available components is called:

(a) chromatography (b) organic synthesis
(c) spectroscopy (d) polymerisation

Answer: (b) organic synthesis — This is how the book defines organic synthesis.

4. Identifying the functional group present in a compound helps to predict its:

(a) price (b) isotopic mass
(c) colour only (d) properties

Answer: (d) properties — The functional group determines the properties and reactions.

5. The number of functional groups, compared with the millions of organic compounds, is:

(a) exactly one (b) equally large
(c) only a handful (d) more than the compounds

Answer: (c) only a handful — A few functional groups define all the families.

6. Knowledge of functional groups is used to ensure the purity and composition of:

(a) medicines and food additives (b) metals and alloys
(c) rocks and minerals (d) air and water only

Answer: (a) medicines and food additives — The book names medicines and food additives.

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25.1 SYNTHESIS AND RETROSYNTHESIS

7. A good synthesis uses starting materials which are:

(a) cheap and easily available (b) rare and costly
(c) always gaseous (d) always aromatic

Answer: (a) cheap and easily available — This is one of the criteria of a good synthesis.

8. A good synthesis should involve:

(a) only reversible steps (b) the greatest number of steps
(c) the least number of steps (d) no reagents

Answer: (c) the least number of steps — Fewer steps give less loss of product.

9. A good synthesis gives:

(a) no product (b) a low overall yield
(c) many side products (d) a high overall yield

Answer: (d) a high overall yield — High overall yield is a criterion of a good route.

10. Reactions of organic synthesis either bring about FGI or:

(a) ionisation (b) carbon–carbon bond formation
(c) sublimation (d) neutralisation

Answer: (b) carbon–carbon bond formation — These are the two categories given in the book.

11. Carbon–carbon bond formation is important because the target molecule is assembled from:

(a) polymers only
(b) larger molecules
(c) simple and smaller starting materials
(d) ionic salts

Answer: (c) simple and smaller starting materials — This is the basic strategy of synthesis.

12. A primary aliphatic amine is prepared in one step from a haloalkane and:

(a) nitric acid (b) water
(c) hydrogen chloride (d) ammonia

Answer: (d) ammonia — $R–CH_2–Cl + NH_3 \rightarrow R–CH_2–NH_2 + HCl$.

13. The by-product when a chloroalkane reacts with ammonia to give a primary amine is:

(a) HCl (b) H_2O
(c) Cl_2 (d) N_2

Answer: (a) HCl — Hydrogen chloride is eliminated.

14. An alcohol is converted into a chloroalkane by:

(a) NH_3
(b) $SOCl_2$
(c) $NaBH_4$
(d) $K_2Cr_2O_7$

Answer: (b) $SOCl_2$ — Thionyl chloride replaces –OH by –Cl.

15. The haloalkane needed for an amine is made from an alcohol because the alcohol is:

(a) more easily available
(b) more reactive towards ammonia
(c) a gas
(d) an amine

Answer: (a) more easily available — The haloalkane is not easily available.

16. The synthesis of an alkyl amine from an alcohol is completed in:

(a) three steps
(b) one step
(c) two steps
(d) five steps

Answer: (c) two steps — Alcohol → chloride → amine.

17. Retrosynthesis is:

(a) a type of isomerism
(b) the same as synthesis
(c) a purification method
(d) exactly the opposite of synthesis

Answer: (d) exactly the opposite of synthesis — It works backward from the target molecule.

18. Retrosynthesis uses the:

(a) distillation approach
(b) disconnection approach
(c) titration approach
(d) combustion approach

Answer: (b) disconnection approach — Bonds of the target molecule are disconnected.

19. In retrosynthesis, those bonds are disconnected which give:

(a) a considerable structural simplification
(b) a more complex structure
(c) a coloured product
(d) a polymer

Answer: (a) a considerable structural simplification — The aim is simpler, available compounds.

20. The retrosynthetic arrow is:

(a) a double-headed arrow (↔)
(b) an open-ended arrow (⟹)
(c) a half arrow (⇌)
(d) a curly arrow

Answer: (b) an open-ended arrow (⟹) — ⟹ points from the target to its precursors.

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21. The molecule which is to be prepared is called the:

(a) synthetic equivalent (b) synthon
(c) reagent (d) target molecule

Answer: (d) target molecule — It is abbreviated TM.

22. TM stands for:

(a) transition metal (b) total mass
(c) target molecule (d) test material

Answer: (c) target molecule — Abbreviation used in retrosynthesis.

23. The imaginary breaking down of the target molecule into simpler starting materials is:

(a) ionisation (b) interconversion
(c) condensation (d) disconnection

Answer: (d) disconnection — This is the definition of disconnection.

24. Fragments of the target molecule generated by a disconnection are called:

(a) synthons (b) isomers
(c) catalysts (d) monomers

Answer: (a) synthons — Synthons are the building blocks for the synthesis.

25. Synthons are often:

(a) noble gases (b) ionic species
(c) polymers (d) free metals

Answer: (b) ionic species — A disconnected bond gives a cation and an anion.

26. A reagent which carries out the function of a synthon is called a:

(a) retrosynthetic tree (b) target molecule
(c) synthetic equivalent (d) functional group

Answer: (c) synthetic equivalent — The synthon itself is usually unstable.

27. A synthon cannot be used itself because it is usually:

(a) insoluble (b) too heavy
(c) coloured (d) unstable

Answer: (d) unstable — So a stable reagent is used in its place.

28. SE stands for:

(a) starting element (b) synthetic equivalent
(c) simple ester (d) synthon energy

Answer: (b) synthetic equivalent — Abbreviation used in retrosynthesis.

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29. The conversion of one functional group into another to reach the target molecule is:

(a) FGI (b) TM
(c) SE (d) disconnection

Answer: (a) FGI — Functional group interconversion.

30. A diagram showing all possible disconnections and synthons from the target molecule to the starting materials is the:

(a) energy profile (b) phase diagram
(c) retrosynthetic tree (d) flow meter

Answer: (c) retrosynthetic tree — This is the definition of the retrosynthetic tree.

31. In the retrosynthesis of aniline, FGI gives:

(a) benzoic acid (b) phenol
(c) toluene (d) nitrobenzene

Answer: (d) nitrobenzene — $-NH_2$ is changed into $-NO_2$.

32. The synthetic equivalent of the aryl anion is:

(a) aniline (b) nitric acid
(c) benzene (d) phenol

Answer: (c) benzene — Neutralising the anion gives benzene.

25.1 FUNCTIONAL GROUP INTERCONVERSION

33. An alkene is converted into a halogenoalkane by:

(a) KOH (alcoholic) (b) HBr
(c) $NaBH_4$ (d) NH_3

Answer: (b) HBr — Electrophilic addition of HBr.

34. A halogenoalkane is converted into an alkene by:

(a) PBr_3
(b) KOH in water
(c) KOH in alcohol
(d) H_2O, H_2SO_4

Answer: (c) KOH in alcohol — Alcoholic KOH causes elimination.

35. A halogenoalkane is converted into an alcohol by:

(a) $SOCl_2$ (b) KOH (alcoholic)
(c) $LiAlH_4$ (d) KOH (aq)

Answer: (d) KOH (aq) — Aqueous KOH causes substitution.

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36. An alcohol is converted into a bromoalkane by:

(a) PBr_3 (b) KOH (aq)
(c) $NaBH_4$ (d) NH_3

Answer: (a) PBr_3 — Phosphorus tribromide replaces –OH by –Br.

37. An alkene is converted into an alcohol by:

(a) $LiAlH_4$
(b) HBr
(c) PBr_3
(d) H_2O, H_2SO_4

Answer: (d) H_2O, H_2SO_4 — Acid-catalysed hydration.

38. An alcohol is converted into an alkene by:

(a) KOH (aq)
(b) heating with H_2SO_4
(c) $NaBH_4$
(d) $K_2Cr_2O_7$

Answer: (b) heating with H_2SO_4 — Dehydration.

39. A primary alcohol is oxidised to an aldehyde by:

(a) $LiAlH_4$
(b) $NaBH_4$
(c) $K_2Cr_2O_7$, H_2SO_4
(d) PBr_3

Answer: (c) $K_2Cr_2O_7$, H_2SO_4 — Acidified dichromate is the oxidising agent.

40. A carboxylic acid is reduced to an alcohol by:

(a) $LiAlH_4$
(b) $K_2Cr_2O_7$, H_2SO_4
(c) $SOCl_2$
(d) HBr

Answer: (a) $LiAlH_4$ — A strong reducing agent is needed.

41. A carboxylic acid is converted into an ester by:

(a) NH_3
(b) C_2H_5OH, H^+
(c) $SOCl_2$
(d) $LiAlH_4$

Answer: (b) C_2H_5OH, H^+ — Esterification.

42. An acid halide is converted into an amide by:

(a) HBr (b) H_2O
(c) $LiAlH_4$ (d) NH_3

Answer: (d) NH_3 — $CH_3COCl + NH_3 \rightarrow CH_3CONH_2 + HCl$.

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43. An amide is converted into an amine by:

(a) $K_2Cr_2O_7$
(b) $SOCl_2$
(c) $LiAlH_4$
(d) KOH (aq)

Answer: (c) $LiAlH_4$ — Reduction of the amide.

44. A halogenoalkane is converted into an amine by:

(a) NH_3 (b) H_2O
(c) KOH (aq) (d) PBr_3

Answer: (a) NH_3 — Nucleophilic substitution.

45. Benzene is converted into toluene by:

(a) HNO_3, H_2SO_4
(b) CH_3COCl, $AlCl_3$
(c) CH_3Cl, $AlCl_3$
(d) SO_3, H_2SO_4

Answer: (c) CH_3Cl, $AlCl_3$ — Friedel–Crafts alkylation.

46. Benzene is converted into acetophenone by:

(a) CH_3Cl, $AlCl_3$
(b) CH_3COCl, $AlCl_3$
(c) $KMnO_4$, H_2SO_4
(d) Sn, HCl

Answer: (b) CH_3COCl, $AlCl_3$ — Friedel–Crafts acylation.

47. Benzene is converted into nitrobenzene by:

(a) Cl_2, light
(b) SO_3, H_2SO_4
(c) Sn, HCl
(d) HNO_3, H_2SO_4

Answer: (d) HNO_3, H_2SO_4 — Nitration.

48. Nitrobenzene is converted into aniline by:

(a) Sn, HCl
(b) HNO_3, H_2SO_4
(c) $KMnO_4$, H_2SO_4
(d) NaOH

Answer: (a) Sn, HCl — Reduction of the nitro group.

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49. Benzene is converted into benzenesulfonic acid by:

(a) HNO_3, H_2SO_4
(b) SO_3, H_2SO_4
(c) CH_3Cl, $AlCl_3$
(d) Sn, HCl

Answer: (b) SO_3, H_2SO_4 — Sulfonation.

50. Benzenesulfonic acid is converted into phenol by:

(a) CH_3Cl, $AlCl_3$
(b) Sn, HCl
(c) $KMnO_4$, H_2SO_4
(d) (i) NaOH (ii) HCl

Answer: (d) (i) NaOH (ii) HCl — As shown in the aromatic FGI figure.

51. Toluene is converted into benzoic acid by:

(a) Sn, HCl
(b) Cl_2, light
(c) $KMnO_4$, H_2SO_4
(d) $NaBH_4$

Answer: (c) $KMnO_4$, H_2SO_4 — Side-chain oxidation.

52. Toluene is converted into benzyl alcohol by:

(a) (i) Cl_2, light (ii) $KOH_{(aq)}$
(b) $KMnO_4$, H_2SO_4
(c) HNO_3, H_2SO_4
(d) SO_3, H_2SO_4

Answer: (a) (i) Cl_2, light (ii) $KOH_{(aq)}$ — Side-chain chlorination, then hydrolysis.

53. FGI allows the chemist to create the carbon skeleton and also to:

(a) purify the product
(b) measure the yield
(c) select the route to be adopted
(d) find the melting point

Answer: (c) select the route to be adopted — FGI has a crucial role in designing routes.

54. The carbonyl carbon of aldehydes and ketones is attacked readily by:

(a) cations
(b) electrophiles
(c) free radicals only
(d) nucleophiles

Answer: (d) nucleophiles — It carries a partial positive charge.

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25.2 RETROSYNTHESIS AND SYNTHESIS OF COMPOUNDS

55. In the retrosynthesis of ethyl acetate, the bond disconnected is the:

(a) C–O bond (b) C–C bond
(c) C–H bond (d) C=O bond

Answer: (a) C–O bond — The C–O bond of the ester group is broken.

56. The synthetic equivalents for ethyl acetate are:

(a) phenol and carbon dioxide (b) benzene and nitric acid
(c) acetic acid and ethanol (d) methanol and propanoic acid

Answer: (c) acetic acid and ethanol — They react by esterification.

57. The reaction between acetic acid and ethanol is called:

(a) reduction (b) nitration
(c) acylation (d) esterification

Answer: (d) esterification — An ester and water are formed.

58. The catalyst used in the synthesis of ethyl acetate is:

(a) $AlCl_3$ (b) H_2SO_4
(c) Sn (d) H_3PO_4

Answer: (b) H_2SO_4 — Acid catalyst.

59. The other product of esterification is:

(a) water (b) hydrogen chloride
(c) carbon dioxide (d) ammonia

Answer: (a) water — $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$.

60. The first step in the retrosynthesis of aniline is:

(a) oxidation (b) disconnection of a C–C bond
(c) esterification (d) functional group interconversion

Answer: (d) functional group interconversion — $-NH_2$ is changed to $-NO_2$.

61. A nitro group is attached to benzene by:

(a) nucleophilic addition (b) nucleophilic substitution
(c) electrophilic substitution (d) elimination

Answer: (c) electrophilic substitution — The electrophile is NO_2^+.

62. Nitrobenzene is disconnected at the:

(a) C–C bond (b) C–N bond
(c) N–O bond (d) C–H bond

Answer: (b) C–N bond — This gives the aryl anion and nitronium ion.

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63. Disconnection of nitrobenzene gives the aryl anion and the:

(a) hydroxide ion
(b) acylium cation
(c) ethoxide ion
(d) nitronium cation

Answer: (d) nitronium cation — NO_2^+.

64. The synthetic equivalent of the nitronium cation is:

(a) nitric acid
(b) ammonia
(c) nitrobenzene
(d) sodium nitrite

Answer: (a) nitric acid — HNO_3 with H_2SO_4 gives NO_2^+.

65. The formula of the nitronium ion is:

(a) NO_2^-
(b) NO_2^+
(c) NO_3^-
(d) NH_4^+

Answer: (b) NO_2^+ — It is the electrophile in nitration.

66. The number of steps in the synthesis of aniline from benzene is:

(a) three
(b) one
(c) two
(d) four

Answer: (c) two — Nitration, then reduction.

67. Acetophenone is disconnected at the:

(a) C–O bond
(b) C–C bond
(c) C–N bond
(d) C=O bond

Answer: (b) C–C bond — The bond between the ring and the carbonyl carbon.

68. Disconnection of acetophenone gives the aryl anion and the:

(a) methyl anion
(b) nitronium cation
(c) ethoxide ion
(d) acylium cation

Answer: (d) acylium cation — CH_3CO^+.

69. The synthetic equivalent of the acylium cation is:

(a) acyl chloride
(b) ethanol
(c) nitric acid
(d) benzene

Answer: (a) acyl chloride — CH_3COCl.

70. The catalyst in the synthesis of acetophenone from benzene is:

(a) Sn
(b) H_2SO_4
(c) $AlCl_3$
(d) H_3PO_4

Answer: (c) $AlCl_3$ — Friedel–Crafts acylation.

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71. The reaction of benzene with acyl chloride is a:

(a) Friedel–Crafts alkylation (b) Friedel–Crafts acylation
(c) nitration (d) sulfonation

Answer: (b) Friedel–Crafts acylation — An acyl group is introduced.

72. The synthons obtained from ethyl acetate are an acyl cation and:

(a) the chloride ion (b) the aryl anion
(c) the nitronium ion (d) the ethoxide ion

Answer: (d) the ethoxide ion — From the C–O disconnection.

73. The landmark synthesis of cholesterol was achieved in the:

(a) early 1950s (b) early 1900s
(c) 1972 (d) 2020

Answer: (a) early 1950s — By the Robinson and Woodward groups.

74. Cholesterol was synthesised by groups headed by Robinson and:

(a) Kekulé (b) Eschenmoser
(c) Woodward (d) Wöhler

Answer: (c) Woodward — Robinson at Oxford, Woodward at Harvard.

75. Robert Robinson worked at:

(a) Zurich (b) Harvard
(c) MIT (d) Oxford

Answer: (d) Oxford — Woodward worked at Harvard.

76. In phenyl ethanoate, the bond disconnected in retrosynthesis is the:

(a) C–C bond of the ring (b) C–O bond of the ester linkage
(c) C=O bond (d) C–H bond

Answer: (b) C–O bond of the ester linkage — As in other esters.

77. The synthetic equivalents for phenyl ethanoate are:

(a) phenol and ethanoyl chloride (b) benzene and ethanol
(c) aniline and acetic acid (d) toluene and nitric acid

Answer: (a) phenol and ethanoyl chloride — They give the ester and HCl.

25.3 SYNTHESIS OF PHARMACEUTICAL PRODUCTS

78. One of the most important applications of retrosynthesis is in the synthesis of:

(a) cement (b) alloys
(c) pharmaceutical products (d) glass

Answer: (c) pharmaceutical products — Complex medicinal compounds are traced to simple materials.

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79. Paracetamol is disconnected at the:

(a) C–N bond of the amide group (b) C–O bond of the hydroxyl group
(c) C–C bond of the ring (d) O–H bond

Answer: (a) C–N bond of the amide group — This is Step I.

80. The synthetic equivalents obtained from the first disconnection of paracetamol are:

(a) benzene and nitric acid (b) phenol and carbon dioxide
(c) salicylic acid and acetic anhydride (d) 4-aminophenol and acyl chloride

Answer: (d) 4-aminophenol and acyl chloride — They react by nucleophilic addition–elimination.

81. 4-Aminophenol and acyl chloride react through:

(a) electrophilic addition (b) nucleophilic addition–elimination
(c) free-radical substitution (d) elimination only

Answer: (b) nucleophilic addition–elimination — An amide is formed.

82. In the retrosynthesis of paracetamol, the FGI changes 4-aminophenol into:

(a) aniline (b) 4-chlorophenol
(c) salicylic acid (d) 4-nitrophenol

Answer: (d) 4-nitrophenol — p-Nitrophenol is easy to prepare and reduce.

83. 4-Nitrophenol is disconnected at the:

(a) C–N bond (b) C–O bond
(c) C–C bond (d) O–H bond

Answer: (a) C–N bond — This gives the 4-hydroxy aryl anion and nitronium ion.

84. The starting material for the synthesis of paracetamol is:

(a) benzene (b) phenol
(c) toluene (d) aniline

Answer: (b) phenol — It is nitrated first.

85. The first step in the synthesis of paracetamol from phenol is:

(a) acetylation (b) reduction
(c) nitration (d) sulfonation

Answer: (c) nitration — With HNO_3 and H_2SO_4.

86. 4-Nitrophenol is reduced to 4-aminophenol by:

(a) CH_3COCl
(b) $HNO_3 + H_2SO_4$
(c) $KMnO_4$
(d) Sn + HCl

Answer: (d) Sn + HCl — Reduction of the nitro group.

87. The last step in the synthesis of paracetamol is the reaction of 4-aminophenol with:

(a) CH_3COCl (b) CO_2
(c) HNO_3 (d) Sn + HCl

Answer: (a) CH_3COCl — Acetylation of the amino group.

88. The number of steps in the synthesis of paracetamol from phenol is:

(a) one (b) three
(c) two (d) five

Answer: (b) three — Nitration, reduction, acetylation.

89. The functional groups in paracetamol are:

(a) amine and ketone (b) ester and carboxylic acid
(c) amide and hydroxyl (d) nitro and hydroxyl

Answer: (c) amide and hydroxyl — $HO–C_6H_4–NH–CO–CH_3$.

90. The chemical name of aspirin is:

(a) acetylsalicylic acid (b) 4-aminophenol
(c) phenyl ethanoate (d) acetophenone

Answer: (a) acetylsalicylic acid — Given in the book.

91. Aspirin is first disconnected at the:

(a) C–N bond (b) C–O bond of the ester linkage
(c) C–C bond of the ring (d) O–H bond

Answer: (b) C–O bond of the ester linkage — Disconnection-I.

92. The synthetic equivalents from the first disconnection of aspirin are:

(a) phenol and nitric acid (b) 4-aminophenol and acyl chloride
(c) salicylic acid and acetic anhydride (d) benzene and acyl chloride

Answer: (c) salicylic acid and acetic anhydride — Step II.

93. In Disconnection-II of aspirin, the bond broken in salicylic acid is the:

(a) C=O bond (b) O–H bond
(c) C–O bond of the hydroxyl group (d) C–C bond of the carboxylic group

Answer: (d) C–C bond of the carboxylic group — It gives phenol and carbon dioxide.

94. The synthetic equivalents from the second disconnection of aspirin are:

(a) phenol and carbon dioxide (b) benzene and nitric acid
(c) aniline and acyl chloride (d) toluene and oxygen

Answer: (a) phenol and carbon dioxide — Step IV.

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95. Phenol is converted into salicylic acid by:

(a) Sn and HCl
(b) HNO_3 and H_2SO_4
(c) CO_2 at high temperature and pressure, then H_2SO_4
(d) CH_3Cl and $AlCl_3$

Answer: (c) CO_2 at high temperature and pressure, then H_2SO_4 — First step of the aspirin synthesis.

96. Salicylic acid is converted into aspirin by:

(a) nitric acid
(b) acetic anhydride
(c) carbon dioxide
(d) ammonia

Answer: (b) acetic anhydride — With H_3PO_4 and heat.

97. The catalyst used with acetic anhydride in the synthesis of aspirin is:

(a) V_2O_5
(b) $AlCl_3$
(c) Sn
(d) H_3PO_4

Answer: (d) H_3PO_4 — The mixture is heated.

98. The functional groups in aspirin are:

(a) ester and carboxylic acid
(b) amide and hydroxyl
(c) amine and ester
(d) ketone and alcohol

Answer: (a) ester and carboxylic acid — $CH_3COO–C_6H_4–COOH$.

99. The synthon common to the preparation of aspirin and paracetamol is the:

(a) ethoxide ion
(b) nitronium cation
(c) acylium cation
(d) methyl cation

Answer: (c) acylium cation — CH_3CO^+.

100. An FGI is involved in the synthesis of:

(a) aspirin
(b) paracetamol
(c) ethyl acetate
(d) acetophenone

Answer: (b) paracetamol — $–NO_2$ is reduced to $–NH_2$.

101. The by-product in the last step of the aspirin synthesis is:

(a) carbon dioxide
(b) hydrogen chloride
(c) water
(d) acetic acid

Answer: (d) acetic acid — From acetic anhydride.

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25.4 USE OF AI IN DRUG DESIGNING

102. The present methods of developing new drugs are:

(a) very costly and time consuming (b) cheap and quick
(c) fully automatic (d) free of any limits

Answer: (a) very costly and time consuming — This is why AI is being used.

103. A growing problem with existing antibiotics is:

(a) their low price (b) their bright colour
(c) resistance of pathogens (d) their high solubility

Answer: (c) resistance of pathogens — Bacteria are becoming resistant.

104. Halicin was discovered by researchers at:

(a) Oxford (b) MIT
(c) Harvard (d) Cambridge

Answer: (b) MIT — Massachusetts Institute of Technology.

105. The researchers designed a:

(a) chromatography column (b) new microscope
(c) nuclear reactor (d) machine-learning algorithm

Answer: (d) machine-learning algorithm — It looked for features that make molecules antibacterial.

106. The algorithm looked for molecules effective at killing:

(a) E. coli (b) yeast
(c) viruses (d) mosquitoes

Answer: (a) E. coli — A bacterium.

107. The model was applied to 2500 molecules and a set of natural products numbering:

(a) 23 (b) 800
(c) 100 (d) 8000

Answer: (b) 800 — With different structures and bioactivities.

108. The molecule picked out by the model was named:

(a) Paracetamol (b) Aspirin
(c) Penicillin (d) Halicin

Answer: (d) Halicin — A new antibiotic.

109. A different model predicted that Halicin would have:

(a) no antibacterial activity (b) high toxicity to human cells
(c) low toxicity to human cells (d) a sweet taste

Answer: (c) low toxicity to human cells — So it is likely to be safe.

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110. Halicin killed bacteria which were previously:

(a) harmless (b) resistant to treatment
(c) unknown (d) useful

Answer: (b) resistant to treatment — Many of the most problematic strains.

111. After Halicin, the model screened more than:

(a) one billion molecules (b) 100 molecules
(c) 2500 molecules (d) 100 million molecules

Answer: (d) 100 million molecules — From a database.

112. The screening of the database was completed in:

(a) three months (b) three hours
(c) three days (d) three years

Answer: (c) three days — This shows the speed of AI.

113. The number of further molecules selected by the screening was:

(a) 23 (b) 8
(c) 2 (d) 800

Answer: (a) 23 — Structurally different from existing antibiotics.

114. Of the selected molecules, the number showing antibacterial activity was:

(a) two (b) eight
(c) twenty-three (d) one hundred

Answer: (b) eight — Two of them were particularly powerful.

115. The number of particularly powerful molecules found was:

(a) ten (b) eight
(c) twenty-three (d) two

Answer: (d) two — Out of eight active ones.

116. The total synthesis of vitamin B_{12} was achieved in:

(a) 1928 (b) 1950
(c) 1972 (d) 2020

Answer: (c) 1972 — After over 10 years of collaboration.

117. Vitamin B_{12} was synthesised by Woodward and:

(a) Eschenmoser (b) Robinson
(c) Fleming (d) Kekulé

Answer: (a) Eschenmoser — R.B. Woodward and A. Eschenmoser.

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118. The total synthesis of vitamin B_{12} involved over:

(a) 10 steps (b) 100 steps

(c) 23 steps (d) 1000 steps

Answer: (b) 100 steps — It needed 91 post-doctoral researchers and 12 PhD students.

119. The number of post-doctoral researchers needed for the vitamin B_{12} synthesis was:

(a) 100 (b) 12

(c) 23 (d) 91

Answer: (d) 91 — Together with 12 PhD students.

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